Supply Chain Black Box: Hidden Supplier Dependency Discovery, Calibrated Cascading-Risk Prediction and Optimised Recovery for an FMCG Manufacturer

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***Abstract*—A chocolate and malt-beverage manufacturer sees its Tier-1 suppliers but not the Tier-2/3 suppliers they share. We ask whether this hidden structure can be estimated and turned into better recovery decisions. An end-to-end system (i) predicts 30-day supplier disruption with calibrated probabilities, (ii) ranks candidate hidden links, (iii) propagates shocks through a daily material-flow simulator and (iv) converts predictions into cost-minimising actions and a mixed-integer recovery plan. Supplier-level experiments use a documented synthetic ecosystem (500 suppliers, 69 % of links hidden); two real public datasets test the transferable components. Using only observed links understates Tier-2/Tier-3 exposure 2.1×/10.8×. Logistic regression out-ranks the pre-specified XGBoost model (ROC-AUC +0.040, p=0.008); calibration gains are small and not significant; graph neural networks did not beat tabular models. Over 500 paired replications of a predefined nine-scenario design, optimisation-based recovery cuts the simulated impact of a severe cocoa-trader outage from INR 4.72 to 1.16 crore (saving 3.56 crore, 95 % CI 3.25–3.87, Holm p <0.001) but costs more than doing nothing under mild shocks.**

***Index Terms*—supply chain risk, link prediction, graph neural networks, probability calibration, cascading disruption, recovery optimisation, FMCG**

I. Introduction

A. Business Problem

Disruptions are followed by significantly lower long-run stock returns [1]. A malt-beverage jar needs malt, milk powder, cocoa, sugar, premix and PET jars; each Tier-1 input depends on Tier-2 traders that procurement rarely maps, so shocks "ripple" through invisible links [2]. We support two executable decisions: *(D1) each month, which mitigation action to take for each supplier; (D2) when a shock is reported, which alternates receive which quantities, by which mode.*

B. Research Gap, Novelty and Increment

Hidden-link prediction has been shown on real automotive networks [8], [9], GNN risk prediction on the SupplyGraph FMCG benchmark [15], and disruption mitigation in OR/MS [16]; reviews note that AI studies rarely close the loop from prediction to response [3]. *No study we found asks what the hidden links cost in money once a recovery decision is made, nor whether predicted links improve that decision.* Our increment over [8], [9] is to (a) use predicted links downstream—as exposure inputs and MILP availability de-rating—and (b) measure their value against a rule-based policy that sees only the observed graph. Over [15] we add calibration and a cost-based decision rule. **Not claimed:** new learning algorithms, new calibration methods or a new MILP formulation; all components are standard and cited. This is a new, stand-alone project; no code, data or text is reused from earlier work of the authors.

TABLE I
Research map: how the experiments connect (data flow left to right; "Feeds" = where the output is used).

RQ	Exp. (data)	Metric; test	Verdict	Feeds into
1	E1 (synth.)	AUC, ECE; cl. bootstrap	partly	q_i: D1, MILP
2	E2A (synth.)	AP, gap ratio	supported	E_i; graph of C
3	E2B (both)	PR-AUC; 5 seeds	rejected	model choice
–	E3 (DataCo)	AUC; bootstrap	transfer	delay API only
4	S1–S9 (sim.)	CRN; Wilcoxon–Holm	severe only	tiered policy

C. Research Questions and Hypotheses

RQ1/H1: point-in-time features give discriminative 30-day risk, and post-hoc calibration lowers ECE. **RQ2/H2**: the observed network understates upstream exposure, and link prediction ranks hidden links above chance. **RQ3/H3**: message passing over the supply graph improves risk classification over tabular models. **RQ4/H4**: network-aware optimisation (C) has lower expected impact than rule-based (B) and no intervention (A) at every severity. Table I links each question to its experiment, data, test and downstream use.

II. Related Work

Gradient boosting [4] is a strong tabular baseline, and decision use requires calibrated probabilities [5]–[7]. Hidden-link work [8], [9] uses heuristics [10], embeddings [11] and GNNs [12]–[14], which are our comparators; ripple-effect [2] and OR/MS [16] work covers propagation and mitigation. Table II shows that prior work treats these stages separately.

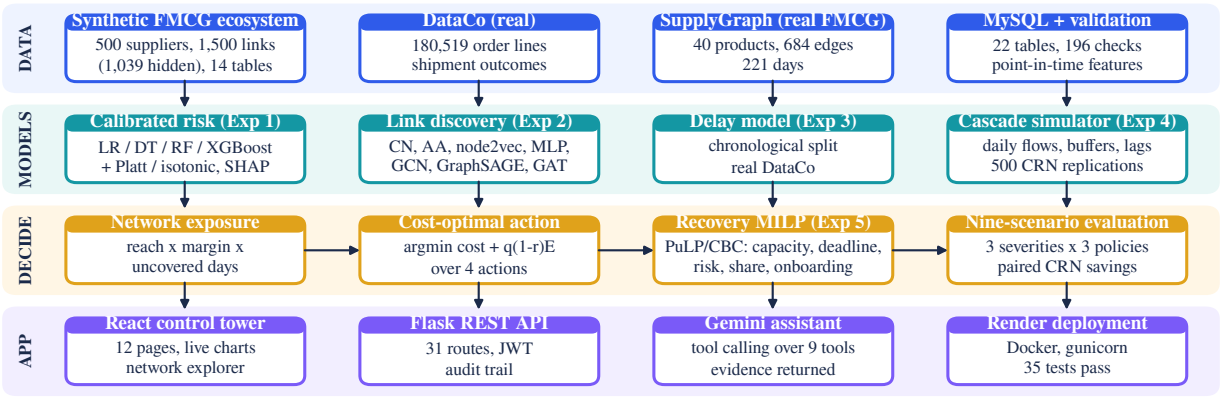


Fig. 1. System architecture. Real and synthetic data feed point-in-time features; calibrated risk, link discovery, delay and cascade models feed an exposure model, a cost-optimal action rule and a recovery MILP; results are served by a Flask API, a React control tower and a tool-calling Gemini assistant, deployed on Render.

TABLE II
SCOPE OF RELATED WORK (✓ PRIMARY FOCUS, ○ PARTIAL / REVIEW).

Work	Hidden links	Risk pred.	Calib-ration	Cas-cade	Recovery opt.	Policy eval.
Brintrup *et al.* [8]	✓					
Kosasih & Brintrup [9]	✓					
Wasi *et al.* [15]		✓				
Baryannis *et al.* [3]		○			○	
Snyder *et al.* [16]				○	✓	○
Ivanov *et al.* [2]		○		✓		
This work	✓	✓	✓	✓	✓	✓

III. METHODOLOGY

A. Data and Provenance

Synthetic ecosystem (fictional, seed 42): 120/200/180 Tier-1/2/3 suppliers, 50 materials, 100 products, two plants, 1,048 disruptions (782 qualifying) in 2024–2025, 4,929 POs, 9,579 shipments. A buyer of material m picks $k \in \{2..6\}$ sources with probability $\propto$ size$^{1.6} e^{-d/2500\,\mathrm{km}}$; Tier-2/3 links are visible with probability 0.45/0.15 (461 of 1,500 observed). The daily hazard combines latent financial risk, utilisation, regional geo-risk, monsoon seasonality, an unobserved frailty and contagion over the *true* network; hidden links serve only as simulator ground truth and for evaluation. **DataCo** [17] (CC BY 4.0, SHA-256 recorded): 180,519 order lines, 7,754 cancelled removed. **SupplyGraph** [15]: 40 products, 684 edges, 221 days. Real and synthetic identities are never merged; all tables pass 196 validation checks.

B. Disruption Prediction and Leakage Control (RQ1)

The prediction timestamp t is 00:00 on the first of each month. For supplier i,

$$y_i(t) = \mathbb{1}\big[\exists e:\ s_e \in (t, t+30\mathrm{d}],\ \ell_e \geq 0.25,\ d_e \geq 3\big], \quad (1)$$

with start s_e, capacity loss ℓ_e and duration d_e. The 25 features use only disruptions reported, and deliveries completed, before t, the monthly risk forecast and *observed* links. Splits: train 2024-04..2025-02 (5,500 rows), one-month embargo, calibration 2025-04..06 (1,500), test 2025-07..12 (3,000). Platt scaling $q = \sigma(a\,\mathrm{logit}\,\hat{p} + b)$ and isotonic regression are fitted on the calibration split only; we report Brier and ECE $= \sum_b \frac{n_b}{N}|\bar{y}_b - \bar{q}_b|$ (10 equal-width bins). XGBoost was fixed as the primary model in the experiment code before any result was seen and was not re-selected on test data.

C. Graph Learning (RQ2, RQ3)

$G = (V, E)$ has 501 nodes with 131 observable attributes. GraphSAGE updates $h_v^{(l+1)} = \mathrm{ReLU}(W_1 h_v^{(l)} + W_2\,\mathrm{mean}_{u\in N(v)} h_u^{(l)})$ and a decoder scores $\sigma(\mathrm{MLP}[z_u \odot z_v, |z_u - z_v|])$. **A1**: observed links split 70/10/20 with test links removed from message passing. **A2**: train on all observed links and rank all 8,863 material-compatible unobserved pairs against 1,039 hidden links; the downstream model is chosen by validation AP only. **E2B** feeds E1 features through GCN/GraphSAGE/GAT on observed, observed+top-500 predicted and oracle graphs, and predicts a 7-day service shortfall on SupplyGraph.

D. Exposure, Cascade Simulation and Decision Rule (D1)

Supplier i's reach r_{im} multiplies dependency shares along paths to the manufacturer, and

$$E_i = \textstyle\sum_m r_{im}\,\mathrm{CM}_m\,\mathbb{E}\big[S\max(0, D - c_m - 10(\tau_i - 1))\big], \quad (2)$$

with daily contribution margin CM_m, cover c_m, tier τ_i and empirical severity/duration (S, D). The simulator propagates availability daily over the true network, $a_b(t) = \min\big(1 - \ell_b(t), \min_i \min(1, A_{bi}(t))\big)$ with $A_{bi}(t) = I_{bi}(t) + \sum_u w_{ub} a_u(t - L_{ub})$, 10-day buffers, lead times L_{ub} and BOM-bottleneck production. The executable action is

$$a^* = \arg\min_a\ C_a + q_i(1 - \rho_a)E_i, \quad (3)$$

over four actions (C_a=0/5k/20k/60k INR, ρ_a=0/0.15/0.45/0.70), Bayes-optimal if q_i is calibrated, with thresholds $q^* = \Delta C/(E_i \Delta\rho)$.

TABLE III

E1: 30-DAY SUPPLIER DISRUPTION PREDICTION ON THE UNTOUCHED TEST SPLIT (3,000 SUPPLIER-MONTHS, PREVALENCE 7.4%). AUCS AFTER PLATT SCALING (RANK-PRESERVING); CALIBRATORS FITTED ON THE CALIBRATION SPLIT ONLY; BOLD = LOWEST ECE PER MODEL; SHADED = PRE-SPECIFIED PRIMARY MODEL. 95% CIS AND TESTS IN TABLE IV.

			Brier		ECE_{10}		
Model	ROC-AUC	PR-AUC	raw	Platt	raw	Platt	Iso.
Logistic reg.	0.666	0.150	0.0669	0.0670	0.0066	**0.0059**	0.0070
Decision Tree	0.571	0.098	0.0704	0.0689	0.0366	**0.0111**	0.0124
Random Forest	0.657	0.140	0.0672	0.0675	**0.0046**	0.0081	0.0063
XGBoost	0.626	0.122	0.0689	0.0680	0.0197	**0.0104**	0.0109

E. Recovery Optimisation and Nine-Scenario Design (RQ4, D2)

For deficits δ_m the MILP (PuLP/CBC) chooses quantities x_{jk} from candidate j by mode k, onboarding z_j and shortage s_m:

$$\min \ \textstyle\sum_{jk} p_j(1 + f_{jk} + hW/2)x_{jk} + \sum_j F z_j + \sum_m \pi_m s_m \tag{4}$$

$$\text{s.t. } \textstyle\sum_{jk} \rho_j(1-q_j)\kappa_j\alpha_j x_{jk} + s_m \geq \delta_m,$$
$$x_{jk} \leq u_j(W - L_{jk})^+, \ \ \textstyle\sum_k x_{jk} \leq 0.6\,\delta_m,$$
$$x_{jk} = 0 \text{ if } q_j > 0.35, \ \ \textstyle\sum_k x_{jk} \leq M z_j,$$

with compatibility κ_j, spare capacity u_j, deadline W, freight f_{jk}, holding rate h, penalty π_m and availability α_j de-rated by estimated dependency on the disrupted suppliers. The shock hits the five largest Tier-2 cocoa-bean traders (19 observed, 25 hidden links) with 25/50/100 % capacity loss for a nominal 60 days. Policies: **A** none; **B** rule-based (observed graph, same-material alternates by reliability, reactive at 10 days of cover); **C** optimisation (observed+predicted graph, MILP at detection plus the same trigger). All policies share network, demand, inventory and economics; the design and the 13 outcome metrics were fixed before any scenario was run. Economics (INR, illustrative): premium 6 %, onboarding 1.5 L, freight 2.5/4 % (expedited 3×), holding 24 %/yr, shortage penalty 15 % of price, interruption 75k per plant-day, recovery management 25k/day.

F. Statistical Analysis

Supplier-months of one supplier are dependent, so E1 uses a *cluster* bootstrap over the 500 suppliers (B=1,000, percentile 95 % CIs, paired differences on the same resample). E3 bootstraps order lines. Scenario policies are compared with common random numbers (replication r uses seed $s_0 + r$ for every policy, 180-day horizon); we report the mean paired saving with a 95 % CI and a Wilcoxon signed-rank test, Holm-adjusted over the nine policy comparisons. GNN results are mean±sd over five seeds. Significance level α=0.05.

IV. RESULTS

A. Calibrated Disruption Prediction (RQ1, H1 partly supported)

Discrimination is modest for these rare, partly contagion-driven events (prevalence 7.4 %; Table III). Logistic regression ranks best (ROC-AUC 0.666, 95 % CI 0.626–0.705; PR-AUC 0.150, twice the base rate) and is significantly better than the pre-specified XGBoost (0.626, CI 0.587–0.664; Table IV); the validation split also favoured it. We keep XGBoost to avoid test-set model selection and test the consequence in a sensitivity run (Sec. IV-E). Platt scaling lowered XGBoost's ECE from 0.020 to 0.010, but the paired reduction is not significant (CI -0.002 to 0.020, p=0.11); for the decision tree it fell from 0.037 to 0.011, while for the already-calibrated random forest Platt scaling *raised* ECE. On the test months the cost rule raised 5 contingency alerts, none followed by a qualifying disruption, and realised cost (118.3 lakh) was close to monitor-only (116.9) and far below review-all (664.3), against a perfect-foresight bound of 82.1. With 3–4 weeks of cover, single-supplier events are mostly absorbed, so value shifts to network-level response.

TABLE IV

STATISTICAL TESTS. E1: SUPPLIER-CLUSTER BOOTSTRAP (B=1,000, 500 SUPPLIERS, 3,000 TEST MONTHS); E3: ORDER-LINE BOOTSTRAP (B=1,000); S: 500 PAIRED CRN REPLICATIONS, 95% CI OF THE MEAN SAVING AND WILCOXON SIGNED-RANK p, HOLM-ADJUSTED OVER NINE COMPARISONS. WIN = SHARE OF REPLICATIONS IN WHICH THE SECOND POLICY IS CHEAPER (L = INR LAKH).

	Comparison	Effect	95% CI	Win	p
E1	LR − XGBoost, ROC-AUC	+0.040	[0.013, 0.068]	–	0.008
E1	LR − XGBoost, PR-AUC	+0.026	[0.002, 0.055]	–	0.044
E1	ECE reduction, XGB raw − Platt	+0.0095	[-0.0018, 0.0197]	–	0.108
E3	XGBoost − mode baseline, ROC-AUC	+0.035	[0.032, 0.039]	–	<0.001
S	Severe: saving of C over A (L)	+356.2	[324.9, 387.4]	80%	<0.001
S	Severe: saving of C over B (L)	+282.0	[252.7, 311.4]	73%	<0.001
S	Moderate: saving of C over A (L)	+16.7	[9.1, 24.3]	17%	<0.001
S	Moderate: saving of B over A (L)	+11.0	[8.7, 13.3]	18%	0.167
S	Mild: saving of C over A (L)	-0.8	[-0.8, -0.8]	0%	<0.001

B. Black-Box Gap and Link Discovery (RQ2 supported; RQ3 rejected)

Observed links understate Tier-2/3 exposure 2.1×/10.8× (Fig. 3). In discovery (Table V) common-neighbour heuristics are uninformative on the layered network; the validation-selected MLP pair model attains AP 0.227 against a base rate of 0.117 (P@100 = 0.32), GraphSAGE 0.168. Graph structure did not improve risk classification (Table VI): even the best GNN variant—including one given the oracle true graph—stays below XGBoost on synthetic data, and on real SupplyGraph logistic regression is best. Predicted links are therefore used as *verification leads*, not as facts.

C. Transfer to Real Shipment Data (DataCo)

On 34,554 chronologically held-out order lines XGBoost reaches ROC-AUC 0.780 (CI 0.775–0.784; PR-AUC 0.858) vs. 0.744 for a shipping-mode baseline, a small but significant gain (+0.035, p <0.001); delay-days regression adds nothing (R^2 0.27 vs. 0.28). DataCo has no supplier identities, so E3 is a transfer check that serves the shipment-lateness API; it does not feed the simulator.

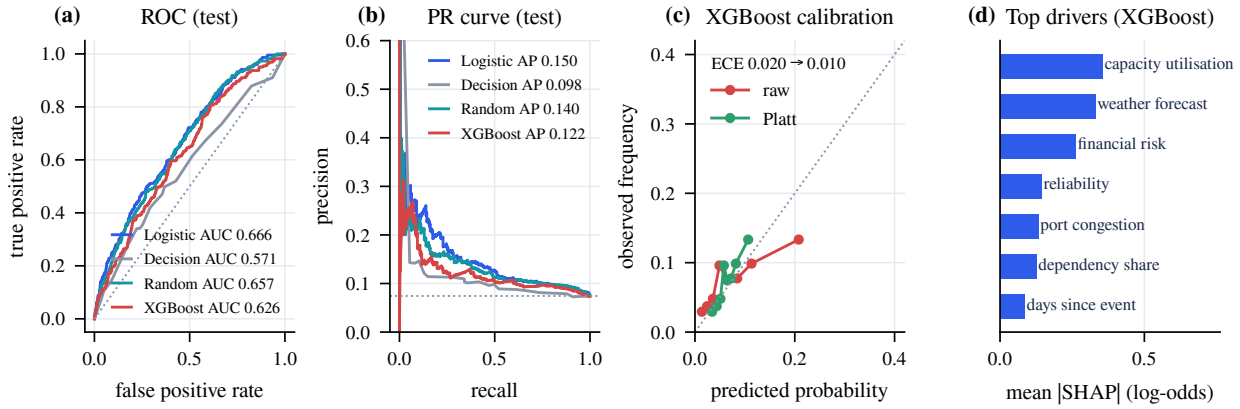


Fig. 2. E1 on the untouched test split (3,000 supplier-months): (a) ROC and (b) precision–recall of the four calibrated models, (c) XGBoost reliability before/after Platt scaling fitted on the calibration split, (d) global SHAP attributions.

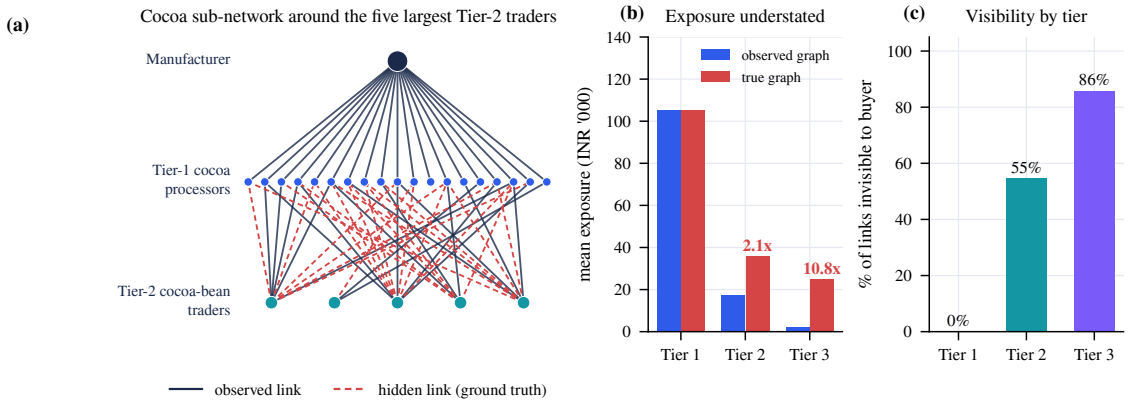


Fig. 3. The black-box gap (synthetic ground truth): (a) cocoa sub-network around the five largest Tier-2 traders—dashed links are invisible to the buyer; (b) mean exposure on the observed vs. true network; (c) share of links invisible by supplier tier.

TABLE V
E2A LINK PREDICTION (SYNTHETIC, 5 SEEDS FOR LEARNED MODELS). A1: HELD-OUT OBSERVED EDGES VS. MATERIAL-COMPATIBLE NEGATIVES. A2: RANKING ALL 8,863 COMPATIBLE UNOBSERVED PAIRS AGAINST 1,039 GROUND-TRUTH HIDDEN EDGES (BASE RATE 0.117).

	A1 (held-out)		A2 (hidden-edge discovery)		
Method	ROC-AUC	AP	AP	P@100	R@500
Common neigh.	0.500	0.500	0.117	0.06	0.042
Adamic-Adar	0.500	0.500	0.117	0.06	0.042
Pref. attach.	0.733	0.763	0.166	0.34	0.120
Node2Vec	0.638	0.609	0.146	0.24	0.084
MLP	0.826±0.017	0.849±0.012	0.227	0.32	0.141
GCN	0.807±0.020	0.837±0.010	0.176	0.25	0.109
GraphSAGE	0.737±0.009	0.784±0.012	0.168	0.29	0.102
GAT	0.796±0.005	0.811±0.018	0.177	0.26	0.103
GraphSAGE+skip	0.792±0.014	0.827±0.006	0.166	0.29	0.102

TABLE VI
E2B: PR-AUC OF GRAPH VS. TABULAR RISK MODELS ON THE TEST SPLIT (PLATT-CALIBRATED; MEAN±SD OVER 5 SEEDS). SYNTHETIC: 30-DAY DISRUPTION, PREVALENCE 7.4%; SUPPLYGRAPH (REAL FMCG): 7-DAY SERVICE SHORTFALL, PREVALENCE 28.8%. THE BEST GNN VARIANT IS REPORTED TO FAVOUR GRAPH MODELS.

Model [graph]	Synthetic	SupplyGraph (real)
Logistic regression (tabular)	–	0.488
XGBoost (tabular)	0.122	0.454
MLP (features, no graph)	0.104±0.011	0.475±0.007
GCN [observed graph]	0.104±0.005	0.459±0.030
GraphSAGE [observed graph]	0.105±0.004	0.461±0.017
GAT [observed graph]	0.100±0.009	0.443±0.040
Best GNN [obs.+predicted links] (GCN)	0.111±0.011	–
Best GNN [oracle true graph] (GAT)	0.108±0.005	–

D. Executable Decision Example (D1, D2)

Table VIII traces one recommendation: calibrated probability, exposure and break-even thresholds select a contingency review as the minimum expected-cost action, and the MILP returns an optimal standby allocation. The API reproduces every value and logs it.

E. Nine-Scenario Evaluation (RQ4, H4 supported for severe shocks only)

Tables IX, IV and Fig. 4. *Severe*: no intervention leaves 734k units unmet (minimum service 64 %, 4.72 crore); rule-based reaches 605k (3.98 crore) because it can exclude only processors with *observed* links to the failed traders. Optimisation pre-books de-rated alternates, cutting unmet demand to 39k units (1.16 crore, full recovery in 88 % of replications); it saves

TABLE IX

NINE-SCENARIO EVALUATION (MEANS OF 500 COMMON-RANDOM-NUMBER REPLICATIONS; SIMULATED INR ON SYNTHETIC DATA). SEVERITY = CAPACITY LOSS OF THE FIVE LARGEST TIER-2 COCOA-BEAN TRADERS FOR A NOMINAL 60 DAYS (LOG-NORMAL). POLICIES: A NONE, B RULE-BASED, C OPTIMISATION. L = LAKH, CR = CRORE; PI = 2.5–97.5TH PERCENTILE OVER REPLICATIONS; FULL REC. = REPLICATIONS WITH ZERO UNMET DEMAND; ALT. UTIL. = SHARE OF ALTERNATES' SPARE CAPACITY USED.

Scen.	Sev.	Pol.	Aff. sup.	Aff. mat.	Unmet (k units)	Inv. short. (L)	Procure. (L)	Transport (L)	Impact, Cr [95% PI]	Recov. d	Alt. util. %	Full rec. %
S1	Mil.	A	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	0	100
S2	Mil.	B	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	1	100
S3	Mil.	C	10.4	2.4	0.0	0.0	4.8	0.12	0.26 [0.01, 0.46]	101	5	100
S4	Mod.	A	17.1	3.9	47.6	2.8	0.0	0.00	0.61 [0.23, 4.54]	123	0	82
S5	Mod.	B	17.1	3.9	29.8	1.9	3.0	0.12	0.51 [0.23, 3.70]	123	16	88
S6	Mod.	C	17.1	3.7	2.6	0.2	86.2	2.95	0.45 [0.38, 0.63]	123	16	97
S7	Sev.	A	19.8	4.0	734.1	46.3	0.0	0.00	4.72 [0.23, 16.10]	127	0	14
S8	Sev.	B	19.8	4.0	605.3	38.9	18.3	0.73	3.98 [0.24, 15.19]	126	42	15
S9	Sev.	C	19.8	3.7	39.2	5.3	461.9	24.20	1.16 [0.93, 3.84]	125	23	88

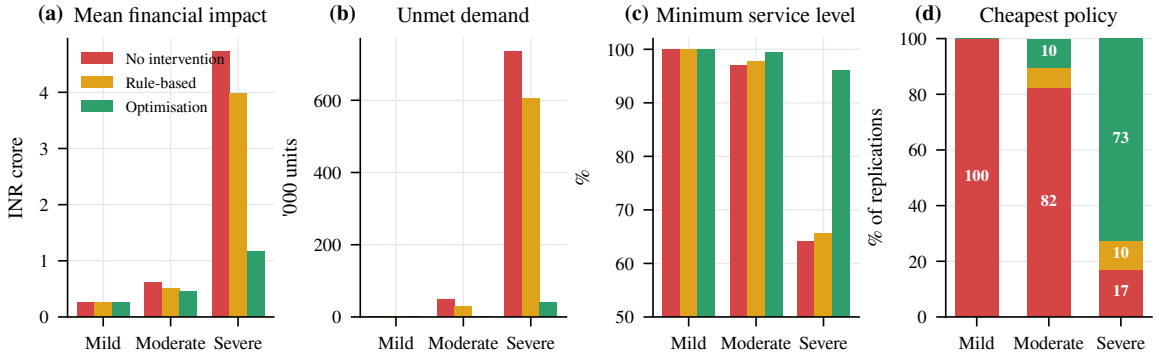


Fig. 4. Nine-scenario outcomes over 500 common-random-number replications (simulated INR on synthetic data): (a) mean financial impact, (b) unmet demand, (c) minimum service level, (d) share of replications in which each policy is cheapest.

TABLE VII

E3 (REAL DATACO): LATE-DELIVERY PREDICTION, CHRONOLOGICAL TEST SPLIT (34,554 ORDER LINES, LATE RATE 57.3%), THRESHOLD 0.5.

Model	ROC-AUC	PR-AUC	F1	Brier
Logistic Regression	0.749	0.839	0.682	0.1878
Random Forest	0.777	0.855	0.710	0.1768
XGBoost	0.780	0.858	0.713	0.1768
Shipping-mode rate	0.744	0.785	0.670	0.1881

TABLE VIII

WORKED DECISION FOR SUPPLIER S0065 (TIER-1, SYNTHETIC) AT t=2025-12-01, PRODUCED END-TO-END BY THE DECISION ENGINE.

Raw / calibrated 30-day risk: 40.8 % / **15.2 %** Exposure E_i: INR 4.41 lakh
Break-even q^*: 7.6 % / 11.3 % / 36.3 % Cover: 24 d
SHAP drivers: capacity utilisation (+0.67), mean delay (90 d) (+0.41), dependency share (+0.30)

Action	Expected cost (INR '000)
Routine monitoring	66.9
Enhanced monitoring	61.9
Contingency review	56.8 ←
Escalate & recover	80.1

Recovery MILP: Optimal, cost INR 15.1k

3.56 crore over A and 2.82 crore over B and is cheaper than A in 80 % and cheapest of all three in 73 % of replications. *Moderate*: mean savings of C (16.7 lakh over A) come from a minority of tail replications—A is cheapest in 82 % of them, and B vs. A is not significant (Holm p=0.17). *Mild*: impact is almost entirely the assumed management overhead, and C costs 0.82 lakh *more* than A in every replication. Severe-mean relative SE is 3.9 %; recovery time is censored in $\leq$3 % of replications. **Sensitivity:** re-running all nine scenarios with logistic-regression risk in place of XGBoost (risk also sets the simulated failure chance of alternates, so all policies shift slightly) gives a severe-C impact of 1.17 crore and a C-over-A saving of 3.67 crore; the RQ4 conclusion does not depend on the risk model.

Key findings

1 Hidden links matter: observed-only exposure is 2.1× (Tier-2) and 10.8× (Tier-3) too low. **2** Simple beats complex: logistic regression > XGBoost $\geq$ GNNs for risk; link prediction is useful only as a ranked lead list. **3** Calibration helps poorly calibrated learners; its gain for XGBoost is not statistically significant. **4** Optimisation pays only for severe shocks (75 % lower impact); for mild shocks it over-hedges ⇒ a tiered policy.

V. DISCUSSION

Relevance and significance. The system gives an auditable action and allocation (Table VIII, Fig. 5); value concentrates in rare severe shocks, supporting a tiered policy—monitor routinely, optimise only for large reported shocks. **Originality.** The increment is measuring the monetary value of hidden-link estimates inside a recovery decision (Sec. I-B); the negative GNN and calibration results are reported as findings, not

(a) Executive dashboard (b) Network explorer (c) Disruption simulator (d) Gemini assistant

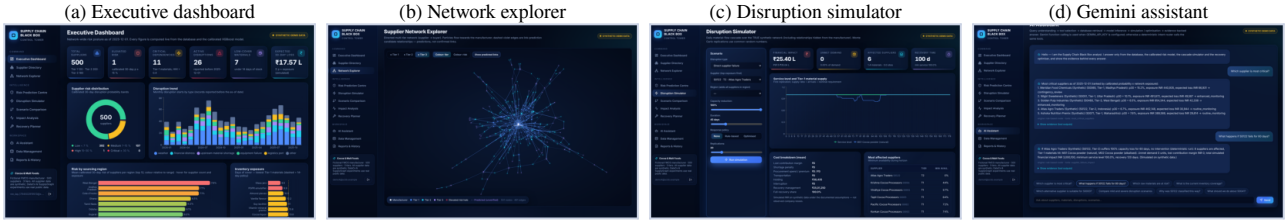


Fig. 5. Deployed application (React control tower, Flask API, MySQL/SQLite, Gemini tool-calling assistant; 35 automated tests pass, all pages browser-tested): (a) executive dashboard, (b) network explorer with predicted links, (c) disruption simulator, (d) evidence-backed assistant.

hidden. **Threats to validity.** *Internal:* the authors designed the generator, so a model matched to it could look better than it would on real data; we mitigate this with leakage controls, fixed seeds and two real datasets, but cannot remove it. *Construct:* the financial impact depends on illustrative unit costs; the 25k/day overhead drives the mild results. *External:* supplier-level results are synthetic, and real-network transfer is untested beyond DataCo and SupplyGraph. *Statistical:* E1 has 223 test positives, so CIs are wide; scenario CIs reflect simulation noise only, not parameter uncertainty.

VI. Research Integrity, AI Use and Reproducibility

Integrity. Nothing was tuned on test data; every table, figure and number here is regenerated from saved result files by script, and 61 automated checks recompute the reported metrics. Disclosed deviations: (i) three simulator corrections were made after initial scenario runs (inventory storage cap that discarded deliveries, policy C excluding rather than de-rating suspect alternates, horizon extended to 180 days to limit censoring); all results use the corrected code; (ii) the statistical tests and the sensitivity run were added after review of the first draft. **AI use declaration.** Claude Code (Anthropic) was used extensively: under the authors' direction it wrote most of the software, ran the experiments and drafted and edited this paper's text. The authors defined the problem, scope, datasets and evaluation requirements, reviewed the design and outputs, and checked every reported number against the logged result files. Google Gemini runs only inside the application's assistant and may phrase answers solely from tool outputs. The authors take full responsibility for the content. **Reproducibility.** Code, data, configurations and logged runs: https://github.com/SamarthKamat/AIML-PROJECT; live demo: https://supply-chain-black-box.onrender.com. `bash run_all.sh` downloads both real datasets at pinned commits with hash checks, regenerates data (seed 42), runs all experiments (GNN seeds 0–4), the statistics and checks, and rebuilds this paper's tables and figures. Software versions are pinned in the repository.

VII. Conclusion

RQ1 partly: risk ranking is modest, logistic regression is best and calibration helps poorly calibrated learners. **RQ2** yes: hidden links multiply upstream exposure and can be ranked at about twice the base rate. **RQ3** no. **RQ4** for severe shocks only (75 % lower impact). Next: validate on real supplier maps (bills of lading, UN Comtrade cocoa flows) and learn rolling-horizon policies that avoid over-hedging.

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